Possibilistic inferential models, properties, and Monte Carlo computations[1]

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[1]Based on: arXiv:2507.09007, arXiv:2501.10585, arXiv:2503.19748

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Introduction

> *Perhaps the most important unresolved problem in statistical inference is the use of Bayes theorem in the absence of prior information.* —Efron

- No prior dist'n represents "absence of prior information"
- So, Bayesian updating of fake priors can't be *right*
- Fisher's fiducial inference apparently also isn't *right*
- What (most of) these failed attempts have in common is that they aim to quantify uncertainty with probability
- Why probabilistic UQ? It's not the only game in town!

Intro, cont.

> *Inferential models are one of the original statistical innovations of the 2010s...* —Cui & Hannig

- A possibilistic IM is a mapping from data, model, etc to a "possibility distribution" for inference
- Mathematical form of IM output: *imprecise probability*
- Imprecision is warranted, but is unfamiliar to statisticians
- Can overcome unfamiliarity with computational strategies
- But computation of imprecise probs can be difficult...
- *Question:* Approximate IM's output by a precise prob?

This talk

- Outline:
  - background: possibilistic IMs & properties
  - probabilistic approximations of possibilistic IMs
  - computation / examples
  - conclusions, etc

- Setup & notation:
  - data $Y \in \mathbb{Y}$, observed value y
  - model $\{\mathsf{P}_\theta : \theta \in \mathbb{T}\}$, uncertain true value is Θ
  - vacuous prior info about Θ
- Goal is inference/UQ about Θ, given $Y = y$

> *Statisticians want numerical measures of the degree to which data support hypotheses.* —Hacking

Possibilistic IMs

Definition — *inferential model* (IM).

A model for quantifying uncertainty about Θ given y, model, etc:

- to every $H \subseteq \mathbb{T}$, assign a pair of numbers

$$\underline{\Pi}_y(H) \quad \text{and} \quad \overline{\Pi}_y(H)$$

- y-dependent measures of $\underline{\text{support for}}$ and $\overline{\text{plausibility of}}$ H

- Representation of knowledge
- Distinct lower and upper prob's implies *imprecision*
- Precise knowledge means $\underline{\Pi}_y \equiv \overline{\Pi}_y$
- Imprecision isn't a shortcoming, flaw, or an unnecessary abstraction — it's necessary for reliability!

Possibilistic IMs, cont.

> *It is unacceptable if a procedure... of representing uncertain knowledge would, if used repeatedly, give systematically misleading conclusions.* —Reid & Cox

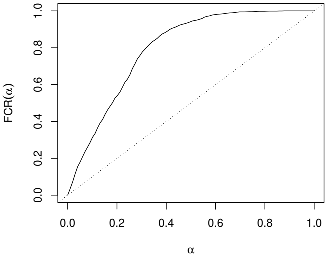
- Let Q_y be a y-dependent probability for Θ, e.g., Bayes
- "Unacceptable" situation:
  - for certain false hypotheses $H \not\ni \Theta$
  - the rv $\mathsf{Q}_Y(H)$ tends to be large under P_Θ
- Define $\text{FCR}_{\mathsf{Q}}(\alpha, H) = \sup_{\theta \notin H} \mathsf{P}_\theta\{\mathsf{Q}_Y(H) > 1 - \alpha\}$

False confidence theorem (Balch, M., Ferson, *Proc Roy Soc* 2019).

Let Q_Y be *any* data-dependent probability for Θ. For any $(\alpha, \beta) \in (0, 1)$, there exists (many) $H \subset \mathbb{T}$ such that $\text{FCR}_{\mathsf{Q}}(\alpha, H) > \beta$

Possibilistic IMs, cont.

- Simple linear regression: $(Y_i \mid x_i) \overset{\text{ind}}{\sim} \mathsf{N}(\beta_0 + \beta_1 x_i, \sigma^2)$
- Covariates are fixed, model parameter $\theta = (\beta_0, \beta_1, \sigma^2)$
- Q_y = Bayes posterior w/ diffuse normal–igamma prior
- Simulation setup:
  - $H = \{(\beta_0, \beta_1, \sigma^2) : \beta_0/\beta_1 < 1\}$ [3]
  - data generated with $\Theta = (0.3, 0.1, 1)$, so H is false



[3] "Non-linearity" in H causes false confidence... (M., `arXiv:2404.16228`)

Possibilistic IMs, cont.

- Reliability requires an imprecise IM
- *Possibility theory*[4] is the simplest imprecise probability
- Similar to probability theory
  - determined by a "density function"
  - but with a different calculus — optimization vs integration
- Specifically:
  - possibility contour $\theta \mapsto \pi_y(\theta)$ with $\sup_\theta \pi_y(\theta) = 1$
  - plausibility/possibility/upper probability

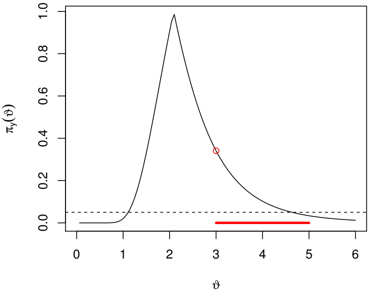
$$\overline{\Pi}_y(H) = \sup_{\theta \in H} \pi_y(\theta), \quad H \subseteq \mathbb{T}$$

  - support/necessity/lower probability $\underline{\Pi}_y(H) = 1 - \overline{\Pi}_y(H^c)$

[4] e.g., Shackle 1961, Zadeh 1975/1978, Dubois & Prade 1988

Possibilistic IMs, cont.

- Illustration of $\pi_y(\theta)$
- Hypothesis $H = [3, 5]$
- $\overline{\Pi}_y(H)$ via optimization
  - "most plausible member"
  - size of H is irrelevant



Possibilistic IMs, cont.

- Imprecise probabilities like $\overline{\Pi}_y$ have a *credal set*,

$$\mathscr{C}(\overline{\Pi}_y) = \{\mathsf{Q}_y \in \mathsf{probs}(\mathbb{T}) : \mathsf{Q}_y(\cdot) \leq \overline{\Pi}_y(\cdot)\}$$

- For possibility measures, there's a nice characterization:

$$\mathsf{Q}_y \in \mathscr{C}(\overline{\Pi}_y) \iff \mathsf{Q}_y\{\pi_y(\Theta) \leq \alpha\} \leq \alpha, \quad \alpha \in [0,1]$$

- Resembles the p-value property...

Possibilistic IMs, cont.

- How to construct a possibilistic IM?
- Workhorse: *probability-to-possibility transform*[5]
- For the present "no-prior" case:
  - relative likelihood, $R(y, \theta) = L_y(\theta) / \sup_\vartheta L_y(\vartheta)$
  - possibility contour

$$\pi_y(\theta) = \mathsf{P}_\theta\{R(Y, \theta) \leq R(y, \theta)\}$$

- Not unfamiliar — p-values!
- However:
  - not significance testing
  - fully conditional & coherent *possibilistic UQ*
  - Fisher: *p-value does not justify any exact probability statement*

[5] M., `arXiv:2211.14567`

Possibilistic IMs, cont.

Strong reliability theorem.

$$\sup_\theta \mathsf{P}_\theta\{\pi_Y(\theta) \le \alpha\} \le \alpha, \quad \alpha \in [0,1]$$

- Possibilistic IM controls false confidence rate, i.e.,

$$\text{FCR}_{\underline{\Pi}}(\alpha, H) \le \alpha \quad \text{for all } H$$

- "Strong" because this isn't just significance testing[6]

$$\sup_\theta \mathsf{P}_\theta\{\underbrace{\overline{\Pi}_Y(H) \le \alpha \text{ for some } H \ni \theta}_{\text{uniformity in } H}\} \le \alpha, \quad \alpha \in [0,1]$$

[6] Cella and M., `arXiv:2304.05740`

Possibilistic IMs, cont.

Corollary.

1. the test "reject H iff $\overline{\Pi}_y(H) \leq \alpha$" satisfies

$$\sup_{\theta \in H} \mathsf{P}_\theta\{\overline{\Pi}_Y(H) \leq \alpha\} \leq \alpha, \quad \alpha \in [0,1], \quad H \subseteq \mathbb{T}$$

2. the region $C_\alpha(y) = \{\theta : \pi_y(\theta) > \alpha\}$ satisfies

$$\sup_{\theta} \mathsf{P}_\theta\{C_\alpha(Y) \not\ni \theta\} \leq \alpha, \quad \alpha \in [0,1]$$

A sort of converse.

For any "valid" frequentist test/confidence procedure, there exists a valid possibilistic IM whose derived procedures are the same or better

Possibilistic IMs, cont.

- Possibilistic IMs are "asymptotically Gaussian"[7]
- Implications:
  - IMs are "efficient" in the classical sense
  - level sets $C_\alpha(y)$ are approximately elliptical...

Possibilistic Bernstein–von Mises theorem.

Simplest-to-state version: under the usual regularity conditions,

$$\sup_{z \in \text{compact}} |\pi_{Y^n}(\hat\theta_{Y^n} + \mathbf{J}_{Y^n}^{-1/2} z) - \gamma(z)| \to 0 \quad \text{in } \mathsf{P}_\Theta\text{-probability},$$

where $\gamma(z) = \mathsf{P}(\|Z\|^2 > \|z\|^2)$ is a standard Gaussian possibility contour

[7] M. and Williams, *IJAR* 2025 (`arXiv:2412.15243`)

Possibilistic IMs, cont.

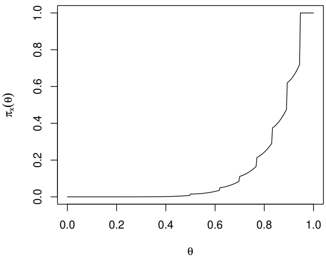
- *More than significance testing!*
- For example, can do formal decision-making[8]
  - $\ell_a(\theta)$ = loss of taking action a when $\Theta = \theta$
  - IM's upper expected loss via Choquet integration

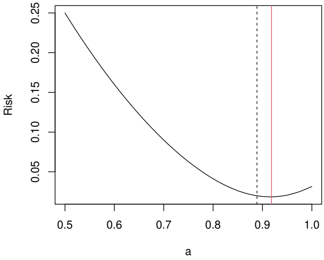
$$\overline{\Pi}_y \ell_a := \int_0^1 \Bigl\{ \sup_{\theta : \pi_y(\theta) > s} \ell_a(\theta) \Bigr\} \, ds$$

  - $\hat{a}(y) = \arg\min_a \overline{\Pi}_y \ell_a$...

[8] M., Prim, and Williams, *IJAR 2026* (arXiv:2112.13247)

- Classical stat problem:
  - all the way back to Hume, Bayes, Laplace, ...
  - observe all successes in n Bernoulli trials
- IM contour (left) and the corresponding risk (right) under squared error loss
- Risk minimizer in red vs Laplace's estimator $\frac{n+1}{n+2}$





Summary so far

- Probabilistic UQ can't guarantee reliability
- Possibilistic UQ can — difference is in the calculus
- New *JASA* review paper (`arXiv:2507.09007`)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Review Article

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Ryan Martin

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Computing possibilistic IMs

- π_y can rarely be evaluated exactly or in closed-form
- Naive Monte Carlo approximation

$$\pi_y(\theta) \approx \frac{1}{M} \sum_{m=1}^{M} 1\{R(Y_m^{(\theta)}, \theta) \leq R(y, \theta)\}, \quad \theta \in \mathbb{T}$$

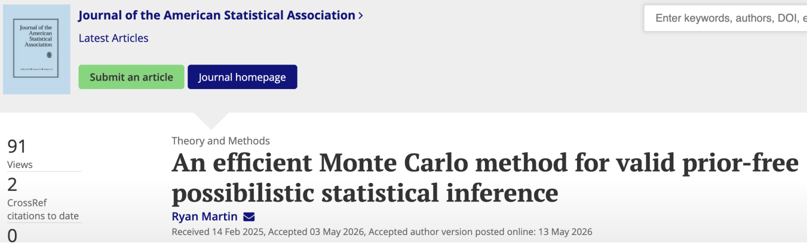
- Conceptually simple, but often expensive to evaluate
- *What about a Bayesian-like strategy?* Sample on the parameter space rather than on the data space
- Not clear how this could/should go:
  - Bayesian-like Monte Carlo evaluates probabilities on $\mathbb{T}$
  - $\overline{\Pi}_y$ isn't a probability on $\mathbb{T}$

Computing possibilistic IMs, cont.

- Suppose we could "approximate" $\overline{\Pi}_y$ by a probability Q_y
- Then those samples yield an approximation of π_y and $\overline{\Pi}_y$, via the probability-to-possibility transform
- This wouldn't require numerous Monte Carlo samples of the data at each parameter value, hence improved efficiency
- Two key questions:
  - how to approximate $\overline{\Pi}_y$ by a probability Q_y?
  - how to sample from the derived Q_y?

Computing possibilistic IMs, cont.

- Key insights in the new *JASA* paper (`arXiv:2501.10585`)
- This probabilistic approximation is treated as an alternative to no-prior Bayes solutions in `arXiv:2503.19748`
- Further refinements are currently in the works



Probabilistic approximation

- *Idea:* approximate $\overline{\Pi}_y$ by a probability in $\mathscr{C}(\overline{\Pi}_y)$...
- Members of $\mathscr{C}(\overline{\Pi}_y)$ are "confidence distributions," i.e.,

$$\mathsf{Q}_y \in \mathscr{C}(\overline{\Pi}_y) \iff \mathsf{Q}_y\{C_\alpha(y)\} \geq 1 - \alpha$$

- An *inner probabilistic approximation* $\mathsf{Q}_y^\star$ of $\overline{\Pi}_y$ is a "most diffuse" member of $\mathscr{C}(\overline{\Pi}_y)$, i.e.,

$$\mathsf{Q}_y^\star\{C_\alpha(y)\} = 1 - \alpha, \quad \alpha \in [0, 1]$$

- Usually exists, but not unique
- What does $\mathsf{Q}_y^\star$ look like?

Probabilistic approximation, cont.

Theorem (M., `arXiv:2501.10585`).

Each member Q_y of $\mathscr{C}(\overline{\Pi}_y)$ can be expressed as

$$\mathsf{Q}_y(\cdot) = \int_0^1 \mathsf{K}_y^{\beta}(\cdot)\,\mathsf{M}(d\beta)$$

- M is a probability stochastically no smaller than $\mathsf{Unif}(0,1)$
- K_y^{β} is a probability fully supported on $C_\beta(y)$ for each $\beta \in [0,1]$

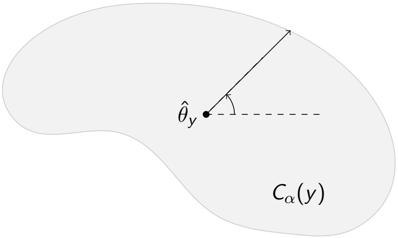
Corollary.

Inner probabilistic approximations $\mathsf{Q}_y^{\star}$ correspond to

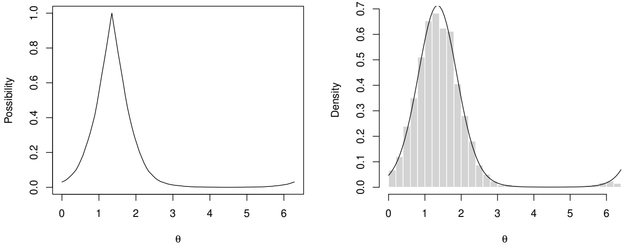
- $\mathsf{M} = \mathsf{Unif}(0,1)$
- K_y^{β} fully supported on $\partial C_\beta(y)$ for each $\beta \in [0,1]$

Probabilistic approximation, cont.

- General proposal: approximate $\overline{\Pi}_y$ by the prob $\mathsf{Q}_y^\star$ with
  - $\mathsf{M} = \mathsf{Unif}(0,1)$
  - $\mathsf{K}_y^\beta = \mathsf{Unif}\{\partial C_\beta(y)\}$ for each $\beta \in [0,1]$
- It's just a suitable uniform mixture of uniforms
- Sampling from $\mathsf{Q}_y^\star$ conceptually straightforward
  - randomly choose level α
  - randomly choose a point on the boundary of $C_\alpha(y)$



- Statistical properties of $\mathsf{Q}_y^\star$ in `arXiv:2503.19748`
- $\mathsf{Q}_y^\star$ agrees with right Haar prior Bayes in invariant models
- Example: von Mises model for directional data
  - invariant to rotations
  - possibilistic IM contour function π_y (left)
  - samples from the inner probabilistic approximation & the right Haar prior Bayes posterior density (right)

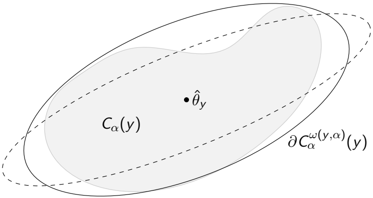


Computational strategy

- Finding a point on $\partial C_\alpha(y)$ is conceptually simple but can be difficult to do
  - boundary may not have a nice shape
  - can only evaluate π_y via Monte Carlo
- Recall: the possibilistic BvM implies that the $\partial C_\alpha(y)$'s are *approximately elliptical*
- If $\partial C_\alpha(y)$ is exactly elliptical, then sampling there is easy
- Actually my first idea was to approximate $\partial C_\alpha(y)$ by a suitable (y, α)-dependent hyper-ellipsoid

Computation, cont.

- Define $C_\alpha^\omega(y) = \{\theta : (\theta - \hat\theta_y)^\top J_y(\omega)^{-1}(\theta - \hat\theta_y) \le \chi^2_{1-\alpha}\}$
- $J_y(\omega)$ is an eigenvalue-adjusted observed Fisher info
- Choose $\omega(y,\alpha)$ such that $C_\alpha(y) \subseteq C_\alpha^{\omega(y,\alpha)}(y)$, snugly
- How to get $\omega(y,\alpha)$...?[9]
- Containment makes approx'n conservative $\implies$ valid



[9] See `arXiv:2501.10585` and `arXiv:2404.19224`

Computation, cont.

Approximate Monte Carlo sampling from $\mathsf{Q}_y^\star$

1. Pre-compute $\omega(y, \alpha_g)$ on a grid $\alpha_1, \ldots, \alpha_G$
2. For each $m = 1, \ldots, M$, do the following
   - draw $A \sim \mathsf{Unif}(0, 1)$
   - set $\omega(A)$ equal to $\omega(y, \alpha_g)$ for α_g closest to A
   - draw $(\Theta_m \mid y, A) \sim \mathsf{Unif}\{\partial C_A^{\omega(A)}(y)\}$
3. Do whatever you usually do with a posterior sample $\Theta_1, \ldots, \Theta_M$

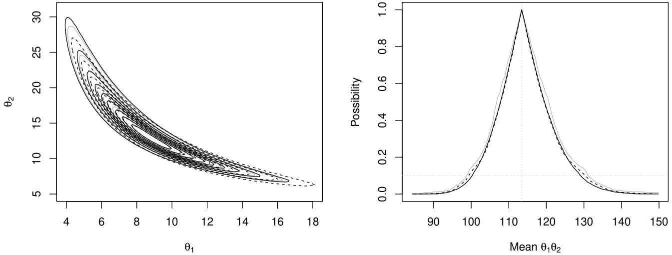
- Step 1 is where all the hard work is...
- Samples aren't Gaussian because ω depends on α
- Elliptical approximation is often very accurate
- Again, elliptical approximation isn't essential

Computation, cont.

- Two details that I'm going to skip:
  - transform back to possibility...
  - eliminating nuisance parameters...
- Happy to discuss later

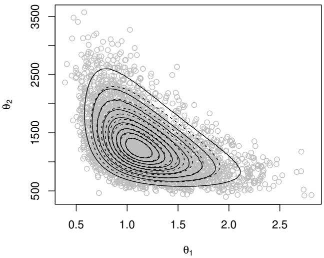
Example: gamma model

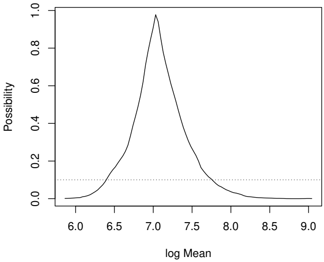
- Real data example, $n = 20$
- Left plot show the joint contour for (shape, scale)
- Right shows marginal contour for the mean



Example: censored Weibull model

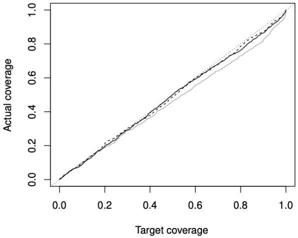
- Real data example, $n = 26$: 14 events, 12 censored
- Left plot shows the joint contour for (shape, scale)
- Right shows marginal contour for the log-mean



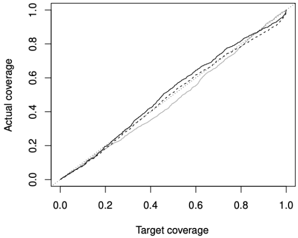


Performance assessments

- Does the approximate possibilistic IM work?
- Check coverage of confidence sets in simulations



(a) Gamma model



(b) Censored Weibull model

Figure 7: Actual coverage probability versus the nominal/target level for three different solutions: Jeffreys prior Bayes (gray), first-order likelihood (dashed), and the proposed approximation to the possibilistic IM (solid). Diagonal dotted line is for reference.

Conclusion

> *Perhaps the most important unresolved problem in statistical inference is the use of Bayes theorem in the absence of prior information.* —Efron

- Maybe the resolution isn't Bayesian or even probabilistic...
- New possibilistic IM framework:
  - likelihood-based
  - familiar and relatively simple
  - achieves strong reliability & efficiency
  - ...
- Computation requires care, but there are cool new ideas

Conclusion, cont.

- At least partially open theory/methods questions:
  - causal inference
  - decision-theory
  - meta-analysis
  - model assessment/selection
  - prediction
- Interesting & unexpected connections to other things:
  - conformal prediction[10]
  - e-values[11]
- My short/medium-term goals:
  - improve computations, develop software
  - application area blitz
  - philosophical grounding

[10] Cella and M., `arXiv:2001.09225` and `arXiv:2112.10234`

[11] M., `arXiv:2303.17425` and `arXiv:2410.01427`

Conclusion, cont.

frequentist		Bayesian
every prior	← some priors →	one prior

- I focused here on the "every prior" case
- But real problems fall in the middle
- e.g., *sparsity* is incomplete prior info[12]
- New-ish n-part series of working papers — *Valid and efficient imprecise-probabilistic inference with partial priors*
- Roughly, $n = 3$ so far[13]

[12] M., Singer, and Williams (`arXiv:2511.09305`)

[13] `arXiv:2203.06703`, `arXiv:2211.14567`, `arXiv:2309.13454`

The end

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- Papers, talks, etc — `www4.stat.ncsu.edu/~rmartin/`
- Questions? — `rgmarti3@ncsu.edu`

Thanks for your attention!